Foundations of Graph attention network

[Graph attention network]

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

1.0 Content Block

h u = σ (1 K ∑ k = 1 K ∑ v ∈ N u α u v k W k x v) {\displaystyle \mathbf {h} _{u}=\sigma \left({\frac {1}{K}}\sum _{k=1}^{K}\sum _{v\in N_{u}}\alpha _{uv}^{k}\mathbf {W} ^{k}\mathbf {x} _{v}\right)}

2.0 Content Block

Water distribution systems can be modelled as graphs, being then a straightforward application of GNN. This kind of algorithm has been applied to water demand forecasting, interconnecting District Metered Areas(DMAs) to improve the forecasting capacity. Other application of this algorithm on water distribution modelling is the development of metamodels.

3.0 Content Block

Local pooling layers Local pooling layers coarsen the graph via downsampling. Subsequently, several learnable local pooling strategies that have been proposed are presented. For each case, the input is the initial graph represented by a matrix X {\displaystyle \mathbf {X} } of node features, and the graph adjacency matrix A {\displaystyle \mathbf {A} } . The output is the new matrix X ′ {\displaystyle \mathbf {X} '} of node features, and the new graph adjacency matrix A ′ {\displaystyle \mathbf {A} '} .

4.0 Content Block

Attention in Machine Learning is a technique that mimics cognitive attention. In the context of learning on graphs, the attention coefficient α u v {\displaystyle \alpha _{uv}} measures how important node v {\displaystyle v} is to node u {\displaystyle u} . Normalized attention coefficients are computed as follows:

5.0 Content Block

where H {\displaystyle \mathbf {H} } is the matrix of node representations h u {\displaystyle \mathbf {h} _{u}} , X {\displaystyle \mathbf {X} } is the matrix of node features x u {\displaystyle \mathbf {x} _{u}} , σ (⋅) {\displaystyle \sigma (\cdot)} is an activation function (e.g., ReLU), A ~ {\displaystyle {\tilde {\mathbf {A} }}} is the graph adjacency matrix with the addition of self-loops, D ~ {\displaystyle {\tilde {\mathbf {D} }}} is the graph degree matrix with the addition of self-loops, and Θ {\displaystyle \mathbf {\Theta } } is a matrix of trainable parameters. In particular, let A {\displaystyle \mathbf {A} } be the graph adjacency matrix: then, one can define A ~ = A + I {\displaystyle {\tilde {\mathbf {A} }}=\mathbf {A} +\mathbf {I} } and D ~ i i = ∑ j ∈ V A ~ i j {\displaystyle {\tilde {\mathbf {D} }}_{ii}=\sum _{j\in V}{\tilde {A}}_{ij}} , where I {\displaystyle \mathbf {I} } denotes the identity matrix. This normalization ensures that the eigenvalues of D ~ − 1 2 A ~ D ~ − 1 2 {\displaystyle {\tilde {\mathbf {D} }}^{-{\frac {1}{2}}}{\tilde {\mathbf {A} }}{\tilde {\mathbf {D} }}^{-{\frac {1}{2}}}} are bounded in the range [0 , 1] {\displaystyle [0,1]} , avoiding numerical instabilities and exploding/vanishing gradients. A limitation of GCNs is that they do not allow multidimensional edge features e u v {\displaystyle \mathbf {e} _{uv}} . It is however possible to associate scalar weights w u v {\displaystyle w_{uv}} to each edge by imposing A u v = w u v {\displaystyle A_{uv}=w_{uv}} , i.e., by setting each nonzero entry in the adjacency matrix equal to the weight of the corresponding edge.

6.0 Content Block

Graph attention network The graph attention network (GAT) was introduced by Petar Veli■kovi■ et al. in 2018. A graph attention network is a combination of a GNN and an attention layer. The implementation of attention layer in graphical neural networks helps provide attention or focus to the important information from the data instead of focusing on the whole data. A multi-head GAT layer can be expressed as follows:

7.0 Content Block

h u (l + 1) = GRU (m u (l + 1) , h u (l)) {\displaystyle \mathbf {h} _{u}^{(l+1)}={\text{GRU}}(\mathbf {m} _{u}^{(l+1)},\mathbf {h} _{u}^{(l)})}

8.0 Content Block

where a {\displaystyle \mathbf {a} } is a vector of learnable weights, · T {\displaystyle \cdot ^{T}} indicates transposition, e u v {\displaystyle \mathbf {e} _{uv}} are the edge features (if present), and LeakyReLU {\displaystyle {\text{LeakyReLU}}} is a modified ReLU activation function. Attention coefficients are then normalized via softmax to make them easily comparable across different nodes. A GCN can be seen as a special case of a GAT where attention coefficients are not learnable, but fixed and equal to the edge weights w u v {\displaystyle w_{uv}} .

9.0 Content Block

To represent an image as a graph structure, the image is first divided into multiple patches, each of which is treated as a node in the graph. Edges are then formed by connecting each node to its nearest neighbors based on spatial or feature similarity. This graph-based representation enables the application of graph learning models to visual tasks. The relational structure helps to enhance feature extraction and improve performance on image understanding.

10.0 Content Block

where i = top k (y) {\displaystyle \mathbf {i} ={\text{top}}_{k}(\mathbf {y})} is the subset of nodes with the top-k highest projection scores, ■ {\displaystyle \odot } denotes element-wise matrix multiplication. The self-attention pooling layer can be seen as an extension of the top-k pooling layer. Differently from top-k pooling, the self-attention scores computed in self-attention pooling account both for the graph features and the graph topology.

11.0 Content Block

Gated graph sequence neural network The gated graph sequence neural network (GGS-NN) was introduced by Yujia Li et al. in 2015. The GGS-NN extends the GNN formulation by Scarselli et al. to output sequences. The message passing framework is implemented as an update rule to a gated recurrent unit (GRU) cell. A GGS-NN can be expressed as follows:

12.0 Content Block

When viewed as a graph, a network of computers can be analyzed with GNNs for anomaly detection. Anomalies within provenance graphs often correlate to malicious activity within the network. GNNs have been used to identify these anomalies on individual nodes and within paths to detect malicious processes, or on the edge level to detect lateral movement.

13.0 Content Block

where K {\displaystyle K} is the number of attention heads, ■ {\displaystyle {\Big \Vert }} denotes vector concatenation, σ (·) {\displaystyle \sigma (\cdot)} is an activation function (e.g., ReLU), N u {\displaystyle N_{u}} is the set of immediate neighbor nodes of node u {\displaystyle u} , including node u {\displaystyle u} itself, α u v k {\displaystyle \alpha _{uv}^{k}} are attention coefficients for the k {\displaystyle k} -th attention head, and W k {\displaystyle W^{k}} is a matrix of trainable parameters for the k {\displaystyle k} -th attention head. For the final GAT layer, the outputs from each attention head are averaged before the application of the activation function. Formally, the final GAT layer can be written as: